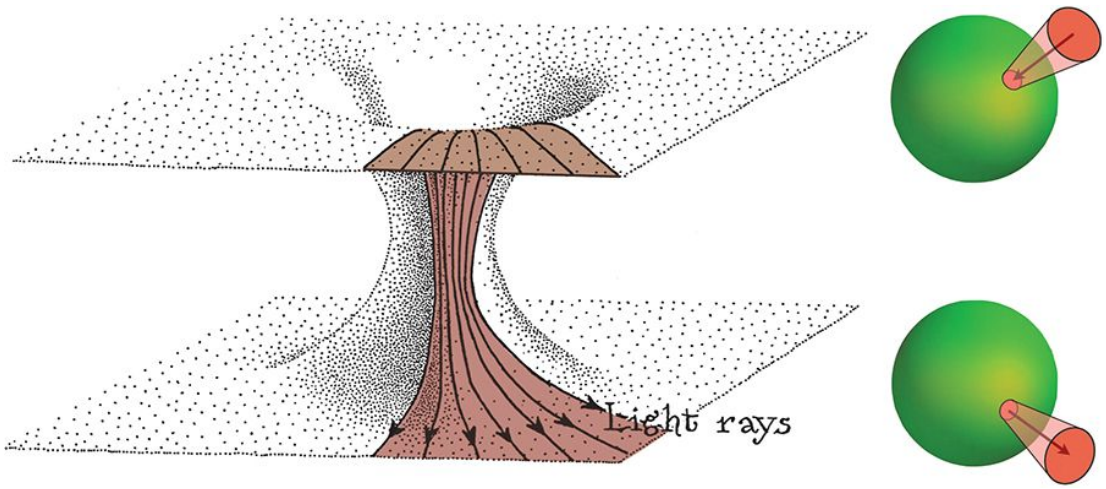


Fig. 14.4. A radial light beam traveling through a spherical, traversable wormhole. *Left*: As seen from the bulk with one space dimension removed. *Right*: As seen in our universe. *[Adapted from a drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*

Now, gravitating bodies such as the Sun or a black hole bend rays inward (Figure 14.5). They can't bend rays outward. To bend light rays outward, a body must have negative mass (or equivalently, negative energy; recall Einstein's equivalence of mass and energy). From this fundamental fact, I concluded that any traversable, spherical wormhole must be threaded by some sort of material that has negative energy. At least the material's energy must be negative as seen by the light beam, or by anything or anyone else that travels through the wormhole at nearly the speed of light.[27] I call such material "exotic matter." (I later learned that, according to Einstein's relativistic laws, *any* wormhole, spherical or not, is traversable only if it is threaded by exotic matter. This follows from a theorem proved in 1975 by Dennis Gannon at the University of California at Davis. Being somewhat illiterate, I was unaware of Gannon's theorem.)

Now, it is an amazing fact that exotic matter *can* exist, thanks to weirdnesses in the laws of quantum physics. Exotic matter has even been made in physicists' laboratories, in *tiny* amounts, between two closely spaced electrically conducting plates. This is called the Casimir